

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 31 — Halogen Compounds

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Reactivity of Halogen Compounds · Halogenoarenes vs Halogenoalkanes vs Acyl Chlorides · CFCs & Ozone
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED

31.1 Relative Reactivity & Hydrolysis Mechanisms

— Comparison of the relative ease of hydrolysis of acyl chlorides (CH_3COCl), alkyl halides ($\text{CH}_3\text{CH}_2\text{X}$)

31.2 Inertness of Halogenoarenes

— Resistance of chlorobenzene to nucleophilic substitution; delocalisation of chlorine lone pair into the ring

31.3 Environmental Impact of CFCs & Ozone Depletion

— Stability of chlorofluorocarbons (CFCs); photolytic homolytic fission by stratospheric UV light generating radicals

31.4 Fluoroalkanes & Bond Energy Considerations

— Extremely high bond dissociation energy of the C-F bond (467 kJ mol^{-1}); chemical inertness and non-flammability

High-Frequency Core Repeats (Q41–Q50)

— The 10 most frequently tested Cambridge Paper 4 questions on Halogen Compounds from the past 10 years

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Comparison of Relative Ease of Hydrolysis of Three Chlorine Compounds — 9701/42/M/J/23/Q8 [6 Marks] **HARD · Syllabus 31.1**

The relative rates of hydrolysis of three chlorine-containing organic compounds are compared in Fig. 1.1:

- Compound A: Ethanoyl chloride, CH_3COCl
- Compound B: 1-chlorobutane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$
- Compound C: Chlorobenzene, $\text{C}_6\text{H}_5\text{Cl}$

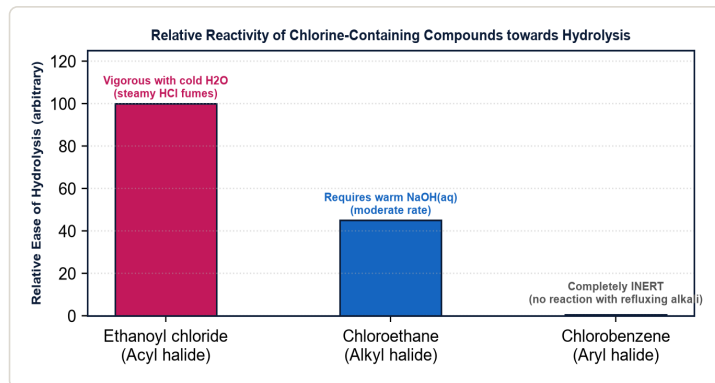


Fig. 1.1: Relative ease of hydrolysis of acyl chloride, alkyl halide, and aryl halide.

- (a) Rank these three compounds in order of decreasing ease of hydrolysis (fastest first). **[1]**

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- (b) Explain why Compound A (ethanoyl chloride) hydrolyses rapidly at room temperature with water alone. **[2]**

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- (c) Explain why Compound C (chlorobenzene) is completely inert to hydrolysis even when boiled with aqueous sodium hydroxide. **[3]**

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2 CFCs and Catalytic Ozone Depletion in the Stratosphere — 9701/41/M/J/23/Q8 [6 Marks] **HARD · Syllabus 31.3**

Dichlorodifluoromethane, CCl_2F_2 (CFC-12), was formerly used as a refrigerant and aerosol propellant. In the stratosphere, it undergoes photolysis by short-wavelength ultraviolet radiation ($\lambda < 220 \text{ nm}$). Data: Bond energies: $\text{C-F} = 467 \text{ kJ mol}^{-1}$; $\text{C-Cl} = 340 \text{ kJ mol}^{-1}$.

- (a) Explain why ultraviolet radiation cleaves the C–Cl bond rather than the C–F bond in CCl_2F_2 , writing an equation for the photolysis. **[2]**

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- (b) Write two equations that show how the chlorine free radical catalyses the destruction of ozone, and construct the overall equation. **[3]**

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- (c) Explain why hydrofluorocarbons (HFCs), such as CH_2FCF_3 , do not cause ozone depletion. [1]
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3 Hydrolysis Rates of 1-Halogenobutanes & Bond Energy Analysis — 9701/42/O/N/23/Q8 [6 Marks]

HARD · Syllabus 31.1

The rate of hydrolysis of primary halogenoalkanes is investigated by adding aqueous silver nitrate in ethanol at 50 °C:

- 1-chlorobutane: white precipitate after several minutes
- 1-bromobutane: cream precipitate after 1 minute
- 1-iodobutane: pale yellow precipitate after 15 seconds
- 1-fluorobutane: no precipitate formed

- (a) Explain the role of ethanol in this test. [1]
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- (b) Write the ionic equation for the precipitation reaction involving 1-iodobutane. [1]
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- (c) Explain the observed trend in the rate of hydrolysis in terms of bond polarity and bond enthalpy, stating which factor dominates. [4]
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4 Mechanistic Pathway: Hydrolysis of Ethanoyl Chloride vs Chloroethane — 9701/41/O/N/23/Q8 [6 Marks]

HARD · Syllabus 31.1

Ethanoyl chloride reacts violently with water at 20 °C, while chloroethane requires heating with aqueous NaOH at 60 °C.

- (a) Outline the nucleophilic addition-elimination mechanism for the reaction of ethanoyl chloride with water, showing curly arrows and intermediates. [3]
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- (b) State the observations when water is added to ethanoyl chloride, identifying the misty fumes evolved. [2]
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- (c) Contrast the mechanism in (a) with the $\text{S}_{\text{N}}2$ mechanism of chloroethane. [1]
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5 Synthesis & Chemical Properties of Halogenoarenes — 9701/42/M/J/22/Q8 [6 Marks]

HARD · Syllabus 31.2

Bromobenzene, C_6H_5Br , is synthesized from benzene by electrophilic substitution.

- (a) State the reagents and catalyst required to synthesize bromobenzene from benzene, and write the overall equation. [2]

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- (b) Describe what is observed when bromobenzene is shaken with aqueous silver nitrate and warmed. [2]

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- (c) Explain why 1-bromobutane gives a cream precipitate with aqueous silver nitrate under the same conditions, while bromobenzene gives no precipitate. [2]

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6 Fluoroalkanes: Polytetrafluoroethene (PTFE) & Thermal Stability — 9701/41/M/J/22/Q8 [6 Marks]

HARD · Syllabus 31.4

Polytetrafluoroethene (PTFE, Teflon) is an addition polymer formed from tetrafluoroethene, $CF_2=CF_2$. PTFE is renowned for its non-stick properties, electrical insulation, and exceptional chemical inertness.

- (a) Draw the repeat unit of PTFE. [1]

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- (b) Explain why PTFE is chemically inert to concentrated acids, concentrated alkalis, and strong oxidising agents even at 250 °C. [3]

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- (c) Explain why PTFE has a very low coefficient of friction (non-stick surface). [2]

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7 Differentiating Three Isomeric Halogen Compounds: C_7H_7Cl — 9701/42/O/N/22/Q8 [6 Marks]

HARD · Syllabus 31.1

Three structural isomers have the molecular formula C_7H_7Cl :

- Isomer 1: (Chloromethyl)benzene, $C_6H_5CH_2Cl$
- Isomer 2: 2-chloromethylbenzene (2-chlorotoluene)
- Isomer 3: 4-chloromethylbenzene (4-chlorotoluene)

- (a) Describe a chemical test that will distinguish Isomer 1 from Isomers 2 and 3, stating reagents, conditions, and observations. [3]

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- (b) Describe a chemical test to confirm that Isomers 2 and 3 possess an alkyl side-chain attached to an aromatic ring. [2]
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- (c) Explain why Isomer 1 hydrolyses faster than 1-chloropropane. [1]
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8 Synthesis & Hydrolysis of Acyl Halides vs Alkyl Halides — 9701/41/O/N/22/Q8 [6 Marks]

HARD · Syllabus 31.1

Carboxylic acids can be converted into acyl chlorides using phosphorus pentachloride, PCl_5 , phosphorus trichloride, PCl_3 , or thionyl chloride, SOCl_2 .

Reaction: $\text{CH}_3\text{COOH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{COCl} + \text{SO}_2 + \text{HCl}$

- (a) Explain why thionyl chloride (SOCl_2) is the preferred laboratory reagent for preparing acyl chlorides. [2]
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- (b) Describe the reaction between ethanoyl chloride and ethanol, naming the organic product and functional group formed. [2]
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- (c) Compare this esterification reaction with the reaction between ethanoic acid and ethanol in terms of speed, reversibility, and catalyst requirements. [2]
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9 Environmental Chemistry: Lifetime of CFCs and the Montreal Protocol — 9701/42/M/J/21/Q8 [6 Marks]

HARD · Syllabus 31.3

CFC-11 (CCl_3F) has an atmospheric lifetime of approximately 50 years and an ozone depletion potential (ODP) of 1.0.

- (a) Explain why CFCs persist unchanged in the troposphere for decades before reaching the stratosphere. [2]
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- (b) Explain how a single chlorine radical can destroy up to 100,000 ozone molecules in the stratosphere. [2]
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- (c) Describe the role of the Montreal Protocol and state two alternative compounds developed to replace CFCs. [2]
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10 Grignard Reagents: Formation and Carbon–Carbon Bond Formation — 9701/41/M/J/21/Q8 [6 Marks]

HARD · Syllabus 31.1

Organomagnesium compounds (Grignard reagents), RMgX, are versatile reagents for extending carbon chains.

Reaction: $\text{CH}_3\text{CH}_2\text{Br} + \text{Mg} \rightarrow \text{CH}_3\text{CH}_2\text{MgBr}$

- (a) State the solvent and conditions required to prepare ethylmagnesium bromide. [2]

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- (b) Explain why ethylmagnesium bromide reacts vigorously with water, writing a balanced chemical equation. [2]

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- (c) Show how ethylmagnesium bromide reacts with methanal, HCHO, followed by acid hydrolysis to produce a primary alcohol, naming the product. [2]

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11 Reactions of Acyl Chlorides: Formation of Amides and Esters — 9701/42/O/N/21/Q8 [6 Marks]

HARD · Syllabus 31.1

Benzoyl chloride, $\text{C}_6\text{H}_5\text{COCl}$, is an aromatic acyl chloride used to prepare esters and amides.

- (a) Write the balanced chemical equation for the reaction of benzoyl chloride with concentrated aqueous ammonia, naming the organic product. [2]

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- (b) Write the balanced chemical equation for the reaction of benzoyl chloride with ethylamine, $\text{CH}_3\text{CH}_2\text{NH}_2$, naming the class of compound formed. [2]

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- (c) Compare the reactivity of benzoyl chloride with ethanoyl chloride towards hydrolysis, explaining the difference. [2]

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12 Polyhalogenated Compounds: Chloroform & Tetrachloromethane Reactions — 9701/41/O/N/21/Q8 [6 Marks]

HARD · Syllabus 31.1

Trichloromethane (chloroform, CHCl_3) and tetrachloromethane (carbon tetrachloride, CCl_4) were historically used as anaesthetics and solvents.

- (a) Explain why CCl_4 is a non-polar molecule even though each C–Cl bond is polar. [2]

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- (b) When exposed to air and light, chloroform is slowly oxidised to highly toxic phosgene, COCl_2 . Write the balanced equation for this oxidation. [2]

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- (c) State why ethanol is often added in small quantities to commercial bottles of chloroform. [2]
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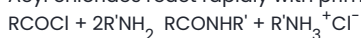
13 Nucleophilic Substitution Kinetics: Primary vs Tertiary Halogenoalkanes — 9701/42/M/J/20/Q8 [6 Marks] HARD · Syllabus 31.1

The kinetics of hydrolysis of 1-bromobutane and 2-bromo-2-methylpropane with NaOH(aq) are investigated.

- (a) Write the rate equation for the hydrolysis of 1-bromobutane and state its mechanism. [2]
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- (b) Write the rate equation for the hydrolysis of 2-bromo-2-methylpropane and state its mechanism. [2]
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- (c) Explain why changing the solvent from 80% ethanol/water to pure water increases the rate of S_N1 hydrolysis of 2-bromo-2-methylpropane. [2]
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14 Nucleophilic Acyl Substitution of Acid Chlorides by Amines — 9701/41/M/J/20/Q8 [6 Marks] HARD · Syllabus 31.1

Acyl chlorides react rapidly with primary amines to form secondary amides:



- (a) Explain why two moles of amine are required per mole of acyl chloride in this reaction. [2]
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- (b) Draw the structural formula of the product formed between propanoyl chloride and phenylamine. [2]
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- (c) State the type of reaction and mechanism occurring. [2]
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15 Fluoroquinolone Antibiotics: Role of Fluorine in Medicinal Chemistry — 9701/42/O/N/19/Q8 [6 Marks] HARD · Syllabus 31.4

Ciprofloxacin is a broad-spectrum fluoroquinolone antibiotic containing a fluorine atom attached directly to an aromatic ring.

- (a) Explain why incorporating a fluorine atom into an aromatic drug increases its metabolic stability in the human body. [2]
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- (b) State how the size of a fluorine atom compares with hydrogen, and explain the concept of 'isosteric replacement'. [2]
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- (c) Explain why replacing an aromatic C-H bond with a C-F bond increases the lipophilicity and membrane permeability of a drug. [2]
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16 Comparison of Halogenoalkane Nucleophilic Substitution Mechanisms — 9701/41/O/N/19/Q8 [6 Marks] **HARD · Syllabus 31.1**

Consider the nucleophilic substitution reactions of 1-iodobutane and 2-iodo-2-methylpropane with cyanide ions, CN^- .

- (a) Draw the transition state for the reaction of 1-iodobutane with CN^- , showing partial bonds and negative charge. [2]
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- (b) Draw the carbocation intermediate for the reaction of 2-iodo-2-methylpropane with CN^- , showing its geometry. [2]
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- (c) State the synthetic utility of converting a halogenoalkane into a nitrile. [2]
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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Comparison of C-Cl Bond Strengths in Chlorobenzene and Chloroethane — 9701/42/M/J/23/Q9 [4 Marks] **MEDIUM · Syllabus 31.2**

The carbon-chlorine bond in chlorobenzene has a bond energy of 400 kJ mol^{-1} , whereas in chloroethane it is 340 kJ mol^{-1} .

- (a) Explain the origin of the partial double-bond character in chlorobenzene. [2]
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- (b) Explain how this partial double-bond character affects bond length and chemical reactivity. [2]
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18 Photolytic Homolytic Fission of CFC-11 — 9701/41/M/J/23/Q9 [4 Marks] **MEDIUM · Syllabus 31.3**

Trichlorofluoromethane, CCl_3F , is broken down by ultraviolet light in the upper atmosphere.

- (a) Write the photochemical equation for the homolytic fission of CCl_3F . [2]
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- (b) Define the term homolytic fission and identify the free radical species produced. [2]
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19 Reaction of Ethanoyl Chloride with Water vs Phenol — 9701/42/O/N/23/Q9 [4 Marks]

MEDIUM · Syllabus 31.1

Ethanoyl chloride reacts with water and with phenol under different conditions.

- (a) Write the equation for the reaction of ethanoyl chloride with water. [2]
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- (b) Write the equation for the reaction of ethanoyl chloride with phenol in the presence of base, naming the ester formed. [2]
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20 Ozone Destruction Propagation Steps by $\text{ClO}^\bullet$ Radicals — 9701/41/O/N/23/Q9 [4 Marks]

MEDIUM · Syllabus 31.3

Chlorine monoxide, $\text{ClO}^\bullet$, is a key intermediate in the catalytic destruction of stratospheric ozone.

- (a) Write the two catalytic propagation steps involving $\text{Cl}^\bullet$ and $\text{ClO}^\bullet$. [2]
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- (b) Explain why chlorine acts as a catalyst in this mechanism. [2]
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21 Comparison of Reactivity of Halogenoalkanes with Aqueous AgNO_3 — 9701/42/M/J/22/Q9 [4 Marks]

MEDIUM · Syllabus 31.1

1-bromobutane and 1-chlorobutane are treated with aqueous silver nitrate in ethanol.

- (a) State which compound forms a precipitate first, giving the colour of the precipitate. [2]
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- (b) Explain this difference in rate by quoting relevant bond enthalpies. [2]
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22 Electrophilic vs Nucleophilic Substitution: Benzene vs Chloroalkanes — 9701/41/M/J/22/Q9 [4 Marks]

MEDIUM · Syllabus 31.2

Different organic functional groups undergo distinct substitution mechanisms.

- (a) Explain why benzene undergoes electrophilic substitution while bromoethane undergoes nucleophilic substitution. [2]
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- (b) Identify the attacking species in each reaction. [2]

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23 Chemical Test to Distinguish Ethanoyl Chloride from Chloroethane — 9701/42/O/N/22/Q9 [4 Marks] MEDIUM · Syllabus 31.1

Ethanoyl chloride and chloroethane are both colourless liquids containing chlorine.

- (a) Describe a simple chemical test using water to distinguish between them, stating observations. [2]

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- (b) Describe what is observed when aqueous silver nitrate is added to the aqueous reaction mixture of each. [2]

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24 Hydrofluorocarbons (HFCs) as Ozone-Friendly Replacements — 9701/41/O/N/22/Q9 [4 Marks] MEDIUM · Syllabus 31.3

HFC-134a (1,1,1,2-tetrafluoroethane) has replaced CFC-12 in domestic refrigerators.

- (a) Draw the structural formula of 1,1,1,2-tetrafluoroethane. [2]

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- (b) Explain why HFCs have zero Ozone Depletion Potential (ODP) but remain an environmental concern. [2]

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25 Reaction of Acyl Chlorides with Ammonia and Primary Amines — 9701/42/M/J/21/Q9 [4 Marks] MEDIUM · Syllabus 31.1

Ethanoyl chloride reacts vigorously with nitrogen bases.

- (a) Write the balanced chemical equation for the reaction of ethanoyl chloride with ammonia. [2]

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- (b) Write the balanced chemical equation for the reaction of ethanoyl chloride with methylamine, naming the organic product. [2]

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26 Inertness of Chlorobenzene towards Ammonia — 9701/41/M/J/21/Q9 [4 Marks] MEDIUM · Syllabus 31.2

When chlorobenzene is heated with concentrated aqueous ammonia, no reaction occurs.

- (a) State two reasons why ammonia cannot displace chlorine from chlorobenzene. [2]

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- (b) State how phenylamine is prepared industrially from benzene. [2]

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27 Reaction of Acyl Chlorides with Alcohols — 9701/42/O/N/21/Q9 [4 Marks]

MEDIUM · Syllabus 31.1

Ethanoyl chloride reacts with propan-1-ol.

- (a) Write the balanced chemical equation, including state symbols. [2]

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- (b) Name the ester product and state one advantage of this reaction over the acid-catalysed reaction using ethanoic acid. [2]

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28 Halogen Free Radical Reactivity: Chlorine vs Bromine in Stratospheric Depletion — 9701/41/O/N/21/Q9 [4 Marks]

MEDIUM · Syllabus 31.3

Halons containing bromine (e.g. CBrClF₂) are also potent ozone depleters.

- (a) Explain why bromine radicals are even more destructive per atom towards ozone than chlorine radicals. [2]

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- (b) Write the propagation steps for bromine-catalysed ozone destruction. [2]

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29 Hydrolysis of 2-Bromopropane: S_N1 vs S_N2 Competition — 9701/42/M/J/20/Q9 [4 Marks]

MEDIUM · Syllabus 31.1

Secondary halogenoalkanes can react via both S_N1 and S_N2 pathways simultaneously.

- (a) Explain why secondary halogenoalkanes undergo both mechanisms. [2]

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- (b) State what experimental kinetic order would be observed if both mechanisms operate at similar rates. [2]

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30 Preparation of Acyl Chlorides Using PCl₅ — 9701/41/M/J/20/Q9 [4 Marks]

MEDIUM · Syllabus 31.1

Phosphorus(V) chloride reacts with ethanoic acid to synthesize ethanoyl chloride.

- (a) Write the balanced chemical equation for this reaction. [2]

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- (b) Identify the two inorganic by-products and describe how ethanoyl chloride is separated from the reaction mixture. [2]

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31 Physical Properties of Fluoroalkanes: Boiling Points vs Alkanes — 9701/42/O/N/19/Q9 [4 Marks]

MEDIUM · Syllabus 31.4

Tetrafluoromethane, CF_4 ($M_r = 88.0$), boils at -128°C , whereas hexane, C_6H_{14} ($M_r = 86.0$), boils at $+69^\circ\text{C}$.

- (a) Explain why CF_4 has an extraordinarily low boiling point despite having a high molecular mass. [2]

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- (b) State the type of intermolecular forces present in liquid CF_4 and liquid hexane. [2]

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32 Comparison of Reactivity: Acyl Chloride vs Alkyl Chloride vs Aryl Chloride — 9701/41/O/N/19/Q9 [4 Marks] MEDIUM · Syllabus 31.1

Order of reactivity towards nucleophilic attack: $\text{CH}_3\text{COCl} > \text{CH}_3\text{CH}_2\text{Cl} > \text{C}_6\text{H}_5\text{Cl}$.

- (a) State which compound reacts most vigorously with water at room temperature, writing the equation. [2]

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- (b) State which compound requires boiling with alcoholic silver nitrate to produce a precipitate. [2]

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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Reagent for Preparing Acyl Chlorides from Carboxylic Acids — 9701/42/M/J/23/Q10 [2 Marks]

EASY · Syllabus 31.1

Acyl chlorides are reactive acid derivatives.

- (a) Name two reagents suitable for converting ethanoic acid into ethanoyl chloride. [2]

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34 Reason for Inertness of Chlorobenzene — 9701/41/M/J/23/Q10 [2 Marks]

EASY · Syllabus 31.2

Aryl halides resist hydrolysis.

- (a) State why the C–Cl bond in chlorobenzene is stronger than in chloroethane. [2]

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35 Definition of a CFC — 9701/42/O/N/23/Q10 [2 Marks]

EASY · Syllabus 31.3

CFCs are synthetic compounds.

- (a) Define the term chlorofluorocarbon (CFC) and give one example formula.

[2]

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36 Bond Cleaved by UV Light in CFCs — 9701/41/O/N/23/Q10 [2 Marks]

EASY · Syllabus 31.3

CFCs break down in the stratosphere.

- (a) State which bond (C-Cl or C-F) is broken by stratospheric UV radiation, and explain why.

[2]

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37 Observation When Ethanoyl Chloride Reacts with Water — 9701/42/M/J/22/Q10 [2 Marks]

EASY · Syllabus 31.1

Acyl chlorides undergo rapid hydrolysis.

- (a) State two visible observations when water is added to ethanoyl chloride.

[2]

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38 Monomer of PTFE — 9701/41/M/J/22/Q10 [2 Marks]

EASY · Syllabus 31.4

PTFE is a commercial fluoropolymer.

- (a) State the name and formula of the monomer used to manufacture PTFE.

[2]

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39 Functional Group Formed by Acyl Chloride + Alcohol — 9701/42/O/N/22/Q10 [2 Marks]

EASY · Syllabus 31.1

Acyl chlorides react with hydroxy compounds.

- (a) Name the organic functional group formed and the inorganic by-product.

[2]

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40 Why HFCs Do Not Deplete Ozone — 9701/41/O/N/22/Q10 [2 Marks]

EASY · Syllabus 31.3

HFCs are used as replacements for CFCs.

- (a) Explain why hydrofluorocarbons do not damage the ozone layer.

[2]

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SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 [CORE REPEAT 1] Comparative Hydrolysis Rates of Chlorine Compounds — 9701/42/M/J/23/Q8 [6 Marks]

HARD · Syllabus 31.1

The relative reactivity of three chlorine compounds towards water is shown in Fig. 41.1.

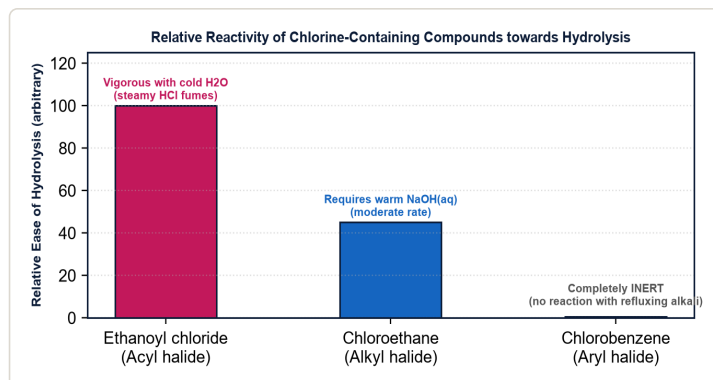


Fig. 41.1: Relative ease of hydrolysis: ethanoyl chloride >> chloroethane >> chlorobenzene.

- (a) Explain why ethanoyl chloride hydrolyses rapidly with water alone at room temperature. [2]

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- (b) Explain why chloroethane requires heating with aqueous sodium hydroxide to undergo hydrolysis. [2]

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- (c) Explain why chlorobenzene is completely unreactive towards hydrolysis. [2]

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42 [CORE REPEAT 2] Complete Catalytic Ozone Destruction Mechanism by CFCs — 9701/41/M/J/23/Q8 [6 Marks]

HARD · Syllabus 31.3

CFC-12, CCl₂F₂, releases chlorine radicals upon photolysis in the stratosphere.

- (a) Write the equation for the initiation step triggered by ultraviolet light. [1]

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- (b) Write the two propagation steps and construct the overall balanced equation for the catalytic cycle. [3]

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- (c) Explain why a single chlorine radical can catalytically destroy many thousands of ozone molecules. [2]

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43 [CORE REPEAT 3] Testing Halogenoalkanes with Aqueous AgNO₃ in Ethanol — 9701/42/O/N/23/Q8 [6 Marks] HARD · Syllabus 31.1

Separate samples of 1-chlorobutane, 1-bromobutane, and 1-iodobutane are warmed with ethanolic AgNO₃(aq).

- (a) State the observations for all three halogenoalkanes, stating precipitate colours. [3]

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- (b) Deduce the order of reactivity of the halogenoalkanes from fastest to slowest. [1]

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- (c) Explain this order of reactivity with reference to bond enthalpies. [2]

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44 [CORE REPEAT 4] Acyl Chloride Reactions: Hydrolysis, Esterification & Amide Formation — 9701/41/O/N/23/Q8 [6 Marks] HARD · Syllabus 31.1

Ethanoyl chloride reacts with water, alcohols, and amines via nucleophilic addition-elimination.

- (a) Write the equation for the reaction of CH₃COCl with water, identifying the misty fumes evolved. [2]

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- (b) Write the equation for the reaction of CH₃COCl with ethanol, naming the organic product. [2]

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- (c) Write the equation for the reaction of CH₃COCl with ammonia, naming the organic product. [2]

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45 [CORE REPEAT 5] Partial Double Bond Character in Chlorobenzene — 9701/42/M/J/22/Q8 [4 Marks] MEDIUM · Syllabus 31.2

Chlorobenzene does not react with aqueous silver nitrate even after prolonged boiling.

- (a) Explain why the C-Cl bond in chlorobenzene does not break easily. [2]

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- (b) Explain why the benzene ring repels incoming hydroxide nucleophiles. [2]

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46 [CORE REPEAT 6] Properties and Uses of Fluoroalkanes and PTFE — 9701/41/M/J/22/Q8 [4 Marks] MEDIUM · Syllabus 31.4

PTFE is an exceptionally unreactive polymer with non-stick properties.

- (a) Explain why the C-F bond is so chemically unreactive. [2]
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- (b) State two commercial uses of PTFE. [2]
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-

47 [CORE REPEAT 7] Distinguishing Benzyl Chloride from Chlorobenzene — 9701/42/O/N/22/Q8 [4 Marks] MEDIUM · Syllabus 31.1

$C_6H_5CH_2Cl$ and C_6H_5Cl are tested with warm ethanolic $AgNO_3$.

- (a) State the observation for (chloromethyl)benzene, $C_6H_5CH_2Cl$. [2]
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-

- (b) State the observation for chlorobenzene, C_6H_5Cl , and explain the difference. [2]
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-

48 [CORE REPEAT 8] Synthesis of Esters from Acyl Chlorides vs Carboxylic Acids — 9701/41/O/N/22/Q8 [4 Marks] MEDIUM · Syllabus 31.1

Esters can be synthesized using either an acyl chloride or a carboxylic acid.

- (a) State two advantages of using an acyl chloride rather than a carboxylic acid to synthesize an ester. [2]
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-

- (b) State one disadvantage or hazard of using an acyl chloride. [2]
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49 [CORE REPEAT 9] Ozone Destruction Equation — 9701/42/M/J/21/Q8 [2 Marks] EASY · Syllabus 31.3

The overall breakdown of stratospheric ozone is catalysed by halogen radicals.

- (a) Write the overall chemical equation for the conversion of ozone into oxygen in the stratosphere. [2]
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50 [CORE REPEAT 10] Acyl Chloride Reagent Choice — 9701/41/M/J/21/Q8 [2 Marks] EASY · Syllabus 31.1

Thionyl chloride is widely used in preparative organic chemistry.

- (a) Write the equation for the reaction of benzoic acid with $SOCl_2$. [2]
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-

Mark Scheme — Topic 31 — Halogen Compounds

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	Compound A (ethanoyl chloride) > Compound B (1-chlorobutane) > Compound C (chlorobenzene) [1].	1
1(b)	The carbonyl carbon is bonded to two strongly electronegative atoms (O and Cl), making it highly electron-deficient (δ^+) [1]; Water readily attacks the carbonyl carbon via an addition-elimination mechanism without needing to break the C-Cl bond in the initial step [1].	2
1(c)	A lone pair of electrons on the chlorine atom overlaps with the delocalised π -electron system of the benzene ring [1]; This imparts partial double bond character to the C-Cl bond, making it significantly shorter, stronger, and harder to break [1]; Furthermore, the incoming OH ⁻ nucleophile is electrostatically repelled by the high π -electron density of the benzene ring [1].	3
2(a)	The C-Cl bond has a significantly lower bond dissociation energy (340 kJ mol ⁻¹ vs 467 kJ mol ⁻¹) and is cleaved homolytically by UV light [1]; CCl ₂ F ₂ → CClF ₂ + Cl• [1].	2
2(b)	Propagation 1: Cl• + O ₃ → ClO• + O ₂ [1]; Propagation 2: ClO• + O → Cl• + O ₂ [1]; Overall: O ₃ + O → 2O ₂ (or 2O ₃ → 3O ₂) [1].	3
2(c)	HFCs contain no chlorine atoms; the C-F bonds are too strong to be broken by stratospheric UV light, so no halogen radicals that destroy ozone are formed [1].	1
3(a)	Ethanol acts as a common solvent (mutual solvent) allowing the water-insoluble halogenoalkane and the aqueous silver nitrate to mix into a single homogeneous solution [1].	1
3(b)	Ag ⁺ (aq) + I ⁻ (aq) → AgI(s) [1].	1
3(c)	Electronegativity decreases from F to I, so C-F is the most polar bond with the largest δ^+ on carbon, which on its own would predict C-F to react fastest [1]; However, bond enthalpy decreases sharply down Group 17: C-F (467) > C-Cl (340) > C-Br (280) > C-I (240 kJ mol ⁻¹) [1]; Bond enthalpy is the overriding / dominant factor governing the rate of hydrolysis [1]; The weaker C-I bond is cleaved much more easily and quickly in the rate-determining step, giving the fastest rate [1].	4
4(a)	Water lone pair attacks carbonyl C δ^+ , C=O π -bond breaks to O ⁻ forming a tetrahedral intermediate [1]; O ⁻ reforms C=O double bond, expelling Cl ⁻ as a leaving group [1]; Loss of H ⁺ from protonated oxygen yields ethanoic acid and HCl [1].	3
4(b)	Vigorous exothermic reaction / bubbling; steamy white / misty fumes of hydrogen chloride (HCl) gas [2].	2
4(c)	Ethanoyl chloride reacts by a two-step addition-elimination mechanism via a tetrahedral intermediate, whereas chloroethane reacts by a one-step concerted S _N 2 mechanism with direct C-Cl bond cleavage in the transition state [1].	1
5(a)	Liquid bromine, Br ₂ , and anhydrous iron(III) bromide, FeBr ₃ (or Fe) [1]; C ₆ H ₆ + Br ₂ → C ₆ H ₅ Br + HBr [1].	2
5(b)	No precipitate is formed / solution remains clear / no reaction [2].	2
5(c)	In 1-bromobutane, the C-Br bond is a standard single bond that undergoes nucleophilic substitution by water to release free Br ⁻ ions, precipitating AgBr [1]; In bromobenzene, p- π orbital overlap delocalises the bromine lone pair into the ring, strengthening the C-Br bond and preventing release of Br ⁻ ions [1].	2
6(a)	[–CF ₂ –CF ₂ –] _n drawn with single C-C bond and open end-bonds [1].	1
6(b)	The C-F bond is exceptionally strong with an enormous bond enthalpy (467 kJ mol ⁻¹) [1]; The carbon backbone is completely sheathed/shielded by a dense, unreactive outer cylinder of highly electronegative fluorine atoms with tightly held non-polarisable electron clouds [1]; Reagents cannot access the carbon skeleton to initiate chemical attack [1].	3
6(c)	Fluorine atoms have very low polarisability, resulting in extremely weak London dispersion intermolecular forces with adjacent surfaces [2].	2
7(a)	Add aqueous ethanol and silver nitrate solution at 50 °C [1]; Isomer 1 ((chloromethyl)benzene) produces an immediate white precipitate of AgCl [1]; Isomers 2 and 3 show no precipitate / remain clear [1].	3
7(b)	Heat with alkaline potassium manganate(VII), KMnO ₄ , then acidify with dilute H ₂ SO ₄ [1]; The purple colour decolourises and a white crystalline precipitate of chlorobenzoic acid forms [1].	2
7(c)	The carbocation intermediate formed in the hydrolysis of (chloromethyl)benzene is a benzylic carbocation (C ₆ H ₅ –C ⁺ H ₂), which is resonance-stabilised by delocalisation of charge into the benzene ring [1].	1

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
8(a)	Both by-products (SO ₂ and HCl) are gases and escape from the reaction mixture [1]; Leaving pure liquid ethanoyl chloride without needing complex separation [1].	2
8(b)	CH ₃ COCl + CH ₃ CH ₂ OH → CH ₃ COOCH ₂ CH ₃ + HCl [1]; Product = ethyl ethanoate, functional group = ester (-COO-) [1].	2
8(c)	With ethanoyl chloride: fast/vigorous at room temperature, irreversible, requires no catalyst [1]; With ethanoic acid: slow equilibrium requiring reflux, reversible, requires concentrated H ₂ SO ₄ catalyst [1].	2
9(a)	CFCs are chemically inert, non-flammable, insoluble in water (not washed out by rain), and resistant to oxidation by hydroxyl radicals in the lower atmosphere [2].	2
9(b)	Chlorine acts as a catalytic chain carrier: in the propagation cycle (Cl• + O ₃ → ClO• + O ₂ ; ClO• + O → Cl• + O ₂), the Cl• radical is regenerated unchanged at the end of each cycle [1]; The cycle repeats thousands of times until a termination reaction occurs (e.g. Cl• + CH ₄ → HCl + •CH ₃) [1].	2
9(c)	International treaty banning production and consumption of ozone-depleting substances [1]; Replacements: hydrofluorocarbons (HFCs, e.g. HFC-134a) and hydrofluoroolefins (HFOs) / hydrocarbons (propane/butane) [1].	2
10(a)	Dry ethoxyethane (dry diethyl ether) under reflux in completely moisture-free conditions [2].	2
10(b)	The C-Mg bond is strongly polarised with a carbanion-like δ^- carbon atom that acts as a powerful Brønsted-Lowry base [1]; CH ₃ CH ₂ MgBr + H ₂ O → CH ₃ CH ₃ (ethane gas) + Mg(OH)Br [1].	2
10(c)	CH ₃ CH ₂ MgBr + HCHO → CH ₃ CH ₂ CH ₂ OMgBr → CH ₃ CH ₂ CH ₂ OH + Mg(OH)Br [1]; Product = propan-1-ol [1].	2
11(a)	C ₆ H ₅ COCl + 2NH ₃ → C ₆ H ₅ CONH ₂ + NH ₄ Cl [1]; Product = benzamide [1].	2
11(b)	C ₆ H ₅ COCl + 2CH ₃ CH ₂ NH ₂ → C ₆ H ₅ CONHCH ₂ CH ₃ + CH ₃ CH ₂ NH ₃ Cl [1]; N-substituted amide (secondary amide) [1].	2
11(c)	Benzoyl chloride hydrolyses more slowly than ethanoyl chloride [1]; Overlap of the carbonyl carbon with the π -electron cloud of the benzene ring reduces the δ^+ charge on the carbonyl carbon, making it less electrophilic towards water [1].	2
12(a)	CCl ₄ has a symmetrical tetrahedral geometry (109.5° bond angles) [1]; The four individual C-Cl bond dipoles cancel each other out completely, giving a zero net molecular dipole moment [1].	2
12(b)	2CHCl ₃ + O ₂ → 2COCl ₂ + 2HCl [2].	2
12(c)	Ethanol reacts with any phosgene formed to produce harmless non-toxic diethyl carbonate: COCl ₂ + 2CH ₃ CH ₂ OH → (CH ₃ CH ₂ O) ₂ CO + 2HCl [2].	2
13(a)	Rate = k[CH ₃ CH ₂ CH ₂ CH ₂ Br][OH ⁻] [1]; S _N 2 mechanism (bimolecular nucleophilic substitution) [1].	2
13(b)	Rate = k[(CH ₃) ₃ CBr] [1]; S _N 1 mechanism (unimolecular nucleophilic substitution) [1].	2
13(c)	Water is a more polar protic solvent than aqueous ethanol [1]; A more polar solvent solvates and stabilises the charged carbocation intermediate (and leaving Br ⁻ ion) in the rate-determining step, lowering activation energy and increasing rate [1].	2
14(a)	The first mole of amine acts as a nucleophile to form the amide [1]; The second mole of amine acts as a base to neutralise the acidic by-product HCl, preventing protonation of the unreacted amine [1].	2
14(b)	N-phenylpropanamide: CH ₃ CH ₂ CONH-C ₆ H ₅ drawn correctly [2].	2
14(c)	Condensation reaction / nucleophilic addition-elimination [2].	2
15(a)	The C-F bond is extremely strong (bond enthalpy 467 kJ mol ⁻¹) and resists enzymatic cleavage by Cytochrome P450 oxidases in the liver [2].	2
15(b)	Fluorine has a van der Waals radius (1.47 Å) very close to that of hydrogen (1.20 Å) [1]; Isosteric replacement: substituting F for H does not alter the steric shape of the molecule, allowing it to fit into the original receptor active site [1].	2
15(c)	Fluorine's high electronegativity polarises the C-F bond, reducing overall water solvation and facilitating passive diffusion across hydrophobic cell membrane lipid bilayers [2].	2
16(a)	Planar arrangement of H, H, propyl groups with partial bonds to incoming :N δ^- and outgoing δ^- I at 180°, brackets with negative charge [2].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
16(b)	Planar trigonal tertiary carbocation $(\text{CH}_3)_3\text{C}^+$ drawn with 120° bond angles and positive charge [2].	2
16(c)	It increases the carbon chain length by one carbon atom [1]; and provides an entry point to synthesize carboxylic acids (via acid hydrolysis) or amines (via reduction with $\text{LiAlH}_4/\text{H}_2$) [1].	2
17(a)	One of the lone pairs of electrons in a p-orbital on the chlorine atom overlaps with the delocalised π -electron system of the benzene ring [2].	2
17(b)	Makes the C-Cl bond shorter and stronger [1]; Resisting nucleophilic attack and making chlorobenzene inert to hydrolysis [1].	2
18(a)	$\text{CCl}_3\text{F} + \text{CCl}_2\text{F} + \text{Cl}^\bullet$ [2].	2
18(b)	The breaking of a covalent bond where each bonded atom takes one of the shared pair of electrons, forming two radicals [1]; Chlorine radical, $\text{Cl}^\bullet$ [1].	2
19(a)	$\text{CH}_3\text{COCl} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{HCl}$ [2].	2
19(b)	$\text{CH}_3\text{COCl} + \text{C}_6\text{H}_5\text{OH} \rightarrow \text{CH}_3\text{COOC}_6\text{H}_5 + \text{HCl}$ (or with NaOH phenyl ethanoate + NaCl + H_2O) [1]; Phenyl ethanoate [1].	2
20(a)	Step 1: $\text{Cl}^\bullet + \text{O}_3 \rightarrow \text{ClO}^\bullet + \text{O}_2$ [1]; Step 2: $\text{ClO}^\bullet + \text{O} \rightarrow \text{Cl}^\bullet + \text{O}_2$ [1].	2
20(b)	The chlorine radical initiates ozone destruction in Step 1 and is regenerated chemically unchanged at the end of Step 2 [1]; It is not consumed overall and accelerates the rate of ozone depletion [1].	2
21(a)	1-bromobutane forms a precipitate first [1]; Cream precipitate (AgBr) [1].	2
21(b)	The C-Br bond enthalpy (280 kJ mol^{-1}) is lower than the C-Cl bond enthalpy (340 kJ mol^{-1}) [1]; Less energy is required to break the C-Br bond in the rate-determining step, leading to faster hydrolysis [1].	2
22(a)	Benzene has an electron-rich delocalised π -system that attracts electron-deficient electrophiles [1]; Bromoethane has an electron-deficient δ^+ carbon bonded to electronegative bromine that attracts electron-rich nucleophiles [1].	2
22(b)	Benzene: Electrophile (e.g. NO_2^+ , Cl^+ , Br^+) [1]; Bromoethane: Nucleophile (e.g. $:\text{OH}^-$, $:\text{CN}^-$, $:\text{NH}_3$) [1].	2
23(a)	Add water at room temperature: ethanoyl chloride reacts violently with effervescence / steamy white fumes of HCl; chloroethane shows no visible reaction / forms an immiscible layer [2].	2
23(b)	Ethanoyl chloride: immediate heavy white precipitate of AgCl [1]; Chloroethane: no precipitate formed at room temperature [1].	2
24(a)	$\text{CF}_3\text{-CH}_2\text{F}$ drawn correctly [2].	2
24(b)	They contain no chlorine atoms to form ozone-destroying $\text{Cl}^\bullet$ radicals ($\text{ODP} = 0$) [1]; However, they are potent greenhouse gases with high Global Warming Potential (GWP) that contribute to climate change [1].	2
25(a)	$\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$ [2].	2
25(b)	$\text{CH}_3\text{COCl} + 2\text{CH}_3\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_3 + \text{CH}_3\text{NH}_3\text{Cl}$ [1]; N-methylethanamide [1].	2
26(a)	1. Partial double bond character of the C-Cl bond due to p- π orbital overlap [1]; 2. Electrostatic repulsion of the lone pair on $:\text{NH}_3$ by the delocalised π -system of the ring [1].	2
26(b)	Nitrate benzene to nitrobenzene (conc. HNO_3 + conc. H_2SO_4 at 55°C) [1]; Reduce nitrobenzene to phenylamine using tin and concentrated hydrochloric acid (Sn + conc. HCl), followed by NaOH [1].	2
27(a)	$\text{CH}_3\text{COCl(l)} + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH(l)} \rightarrow \text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3\text{(l)} + \text{HCl(g)}$ [2].	2
27(b)	Propyl ethanoate [1]; The reaction goes to completion (irreversible) and does not require an acid catalyst or heating [1].	2
28(a)	Bromine radicals react with ozone faster than chlorine radicals [1]; and the reservoir molecules that deactivate bromine (such as HBr) are broken down more rapidly by photolysis, keeping active $\text{Br}^\bullet$ in circulation [1].	2
28(b)	$\text{Br}^\bullet + \text{O}_3 \rightarrow \text{BrO}^\bullet + \text{O}_2$ [1]; $\text{BrO}^\bullet + \text{O} \rightarrow \text{Br}^\bullet + \text{O}_2$ [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
29(a)	They have intermediate steric hindrance (allowing backside attack via S _N 2) [1]; and their secondary carbocations have moderate stability via two +I alkyl groups (allowing ionisation via S _N 1) [1].	2
29(b)	Mixed order kinetics: Rate = k ₁ [2-bromopropane] + k ₂ [2-bromopropane][OH ⁻] [2].	2
30(a)	CH ₃ COOH + PCl ₅ → CH ₃ COCl + POCl ₃ + HCl [2].	2
30(b)	Phosphorus(V) trichloride oxide (POCl ₃) and hydrogen chloride (HCl) [1]; Fractional distillation [1].	2
31(a)	Fluorine electrons are held very tightly by the highly electronegative nucleus, resulting in an extremely low polarisability of the electron cloud [1]; Consequently, the London dispersion forces between CF ₄ molecules are exceptionally weak [1].	2
31(b)	Both possess only London dispersion forces (induced dipole-dipole attractions); hexane has an extended chain with much higher polarisability and much stronger London forces [2].	2
32(a)	CH ₃ COCl (ethanoyl chloride) [1]; CH ₃ COCl + H ₂ O → CH ₃ COOH + HCl [1].	2
32(b)	Chloroethane (CH ₃ CH ₂ Cl) requires heating [2] (chlorobenzene does not react at all even on boiling).	2
33(a)	Thionyl chloride (SOCl ₂) [1]; Phosphorus(V) chloride (PCl ₅) (or PCl ₃) [1].	2
34(a)	Delocalisation of a chlorine lone pair into the benzene π-system [1]; gives the C-Cl bond partial double bond character [1].	2
35(a)	An organic compound containing only chlorine, fluorine, and carbon atoms [1]; CCl ₃ F (or CCl ₂ F ₂) [1].	2
36(a)	The C-Cl bond [1]; it has a much lower bond dissociation energy than the C-F bond [1].	2
37(a)	1. Steamy white / misty fumes evolved [1]; 2. Vigorous bubbling / effervescence / temperature rises [1].	2
38(a)	Tetrafluoroethene [1]; CF ₂ =CF ₂ [1].	2
39(a)	Ester (-COO-) [1]; Hydrogen chloride gas (HCl) [1].	2
40(a)	They contain no chlorine atoms [1]; and therefore cannot generate ozone-depleting chlorine radicals [1].	2
41(a)	The carbonyl carbon is bonded to two electronegative atoms (O and Cl), making it very electron-deficient (δ ⁺) [1]; Water readily attacks the carbonyl carbon via an addition-elimination mechanism without initial C-Cl bond cleavage [1].	2
41(b)	The carbon atom has only a moderate δ ⁺ charge and lacks a π-bond for addition [1]; Reaction proceeds by an S _N 2 mechanism requiring substantial activation energy to break the strong C-Cl single bond (340 kJ mol ⁻¹) in the transition state [1].	2
41(c)	Overlap of a chlorine p-orbital lone pair with the delocalised π-electron system of the benzene ring gives the C-Cl bond partial double-bond character [1]; The bond is significantly stronger, and the high electron density of the aromatic ring electrostatically repels incoming OH ⁻ nucleophiles [1].	2
42(a)	CCl ₂ F ₂ → CClF ₂ + Cl• [1].	1
42(b)	Propagation 1: Cl• + O ₃ → ClO• + O ₂ [1]; Propagation 2: ClO• + O → Cl• + O ₂ [1]; Overall: O ₃ + O → 2O ₂ [1].	3
42(c)	Chlorine radicals act as homogeneous catalysts that are regenerated unchanged at the end of every propagation cycle [1]; The chain reaction continues indefinitely until terminated by rare radical-radical recombination [1].	2
43(a)	1-chlorobutane: white precipitate forms slowly; 1-bromobutane: cream precipitate forms moderately fast; 1-iodobutane: pale yellow precipitate forms rapidly [3].	3
43(b)	1-iodobutane > 1-bromobutane > 1-chlorobutane [1].	1
43(c)	The rate-determining step involves cleavage of the carbon-halogen bond [1]; Bond enthalpy decreases from C-Cl (340) to C-Br (280) to C-I (240 kJ mol ⁻¹); the weakest C-I bond breaks most readily, yielding the fastest rate [1].	2
44(a)	CH ₃ COCl + H ₂ O → CH ₃ COOH + HCl [1]; Misty fumes = hydrogen chloride (HCl) [1].	2
44(b)	CH ₃ COCl + CH ₃ CH ₂ OH → CH ₃ COOCH ₂ CH ₃ + HCl [1]; Ethyl ethanoate [1].	2

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
44(c)	$\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$ [1]; Ethanamide [1].	2
45(a)	A lone pair of electrons on the chlorine atom delocalises into the benzene π -system [1]; imparting partial double-bond character and strengthening the bond [1].	2
45(b)	The delocalised π -electron cloud creates a high negative charge density [1]; which electrostatically repels incoming negatively charged OH^- nucleophiles [1].	2
46(a)	The C-F bond is extraordinarily strong with a very high bond dissociation enthalpy (467 kJ mol^{-1}) [1]; and the small, highly electronegative fluorine atoms tightly shield the carbon backbone [1].	2
46(b)	1. Non-stick coatings for cookware / frying pans [1]; 2. Electrical cable insulation / pipe-seal tape / waterproof breathable fabric (Gore-Tex) [1].	2
47(a)	Immediate white precipitate of AgCl formed [2].	2
47(b)	No precipitate formed [1]; In (chloromethyl)benzene, chlorine is on an aliphatic benzylic carbon where C-Cl hydrolyses readily; in chlorobenzene, chlorine is bonded to the aromatic ring with partial double bond character [1].	2
48(a)	1. The reaction is fast / occurs at room temperature without heating or catalyst [1]; 2. The reaction goes to completion (irreversible) giving much higher yields [1].	2
48(b)	Produces toxic and corrosive gaseous hydrogen chloride (HCl) by-product / reacts violently with moisture [2].	2
49(a)	$2\text{O}_3 \rightarrow 3\text{O}_2$ (or $\text{O}_3 + \text{O} \rightarrow 2\text{O}_2$) [2].	2
50(a)	$\text{C}_6\text{H}_5\text{COOH} + \text{SOCl}_2 \rightarrow \text{C}_6\text{H}_5\text{COCl} + \text{SO}_2 + \text{HCl}$ [2].	2